Bulent Soykan, Ph.D., PMP

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Senior Operations Research (OR) Analyst with 25+ years of experience, possessing an extensive skill-set in statistical analysis, OR, applied mathematics, simulation and modeling, machine learning, programming, strategic planning, and the streamlining of business policies and processes, with PMP certification and a track record of successful project completion and goal achievement.

Research Interests

Operations research & systems analysis, data analytics, simulation modeling, agent-based modeling, simulation-based optimization, scheduling algorithms, machine learning, reinforcement learning, sequential decision-making, metaheuristics, experimental design & analysis, systems engineering, project management, transportation & logistics, combat modeling, wargaming, command & control.

Education

Ph.D. in Engineering Management, Department of Engineering Management & Systems Engineering, Old Dominion University, Norfolk/VA/USA, 4.0 G.P.A., *summa cum laude* *2013-2016*
Dissertation: **"**A Hybrid Tabu/Scatter Search Algorithm for Simulation-Based Optimization of Multi-Objective Runway Operations Scheduling**"**

Ph.D. in Operations Research, Defense Sciences Institute, National Defense University, Ankara/Turkey with 3.778 G.P.A. *2009-2015*
Dissertation: "Solution Algorithms and a Decision Support Framework Proposal for Robust Airline Crew Pairing Problem"

Master's Certificate in Project Management, The George Washington University, Washington, D.C./USA *2013-2015*

M.Sc. in Technology Management (Project Management), Defense Sciences Institute, National Defense University (NDU), Ankara/Turkey with 3.818 G.P.A. *2005-2007*

B.Sc. in Systems Engineering, NDU, Ankara/Turkey with 3.567 G.P.A. *1993-1997*

Work Experience

Defense Projects Manager, *HST Automotive, Manufacturing and Industry Inc.*, Ankara/Turkey *2020-2022*

- Managed multiple multi-national defense projects budgets of $ 50+ million to achieve 25%+ profitability and on-time delivery**.**
- Created project forecasting and staffing plans to mitigate risk and ensure project success. Developed action plans and led meetings with executives to review project status and proposed changes to maintain schedule, budget and quality.
- Developed and implemented a data-driven decision support system that optimize information quality and equip leadership with additional insight leading to 15% improvement in decision-making process efficiency.

Project Manager and Senior OR Analyst, *Ministry of National Defense*, Ankara/Turkey *2016-2020*

- Managed missile & rocket systems acquisition projects (worth $ 30 million) including development of operational and system requirements through operational analysis.
- Achieved recognition for using modeling, optimization, simulation, data analytics and evidence-based approaches to improve project management processes.

Project Manager, *NATO HQ Allied Command Transformation*, Norfolk, VA, USA *2013- 2016*

- Managed a multi-national operational command & control project worth $ 5 million.
- Conducted operational requirement analysis to support continued integrity and quality, to ensure requirements remain valid and applicable.

Project Manager and OR Analyst, *Ministry of National Defense*, Ankara/Turkey *2009- 2013*

- Managed dismounted soldier systems acquisition projects worth $ 20 million.
- Implemented applied mathematics, OR, optimization, scheduling algorithms, data analysis techniques to ensure procured systems adhere to quality standards and processes leading to a cost saving of $ 0.7 million.

Operations Research & Systems Analyst, *Ministry of National Defense*, Turkey *1997- 2009*

- Planned and executed advanced military ORSA projects (worth $5 million) along with 5+ analyst and ensured project implementation to improve operational efficiency by 18%.
- Performed detailed analysis of logistics activities to ensure integration support considerations are captured in planning process and reduced logistics disruptions by %9.
- Served as a deployed ORSA analyst in support of NATO SFOR (Bosnia and Herzegovina) and NATO ISAF (Afghanistan) missions.

Skills & Qualifications

Languages: Turkish - Mother tongue
English - Full professional proficiency (National Language Exam (YDS) 96.25/100
TOEFL PBT - Listening 59, Grammar 60, Reading 61, Total 600
NATO STANAG 6001- Listening 4/4, Speaking 4/4, Reading 4/4, Writing 4/4)
German - Basic knowledge French - Basic knowledge

Computer Skills:

- Experienced user of statistical packages, spreadsheet, database, business graphics, word processing, network applications classes of computer software
- Algebraic modeling software: GAMS, AMPL, IBM ILOG OPL, Gurobi, Pyomo, JuMP
- Statistical modeling, & analysis software: R, SPSS
- Simulation modeling software: SIMAN, ARENA, NetLogo, Vensim, Anylogic
- Programming languages: Python, Julia, MATLAB/Octave, C, C++, VBA, SQL
- Typesetting language: LaTeX
- Data visualization: Tableau, Power BI, ggplot2, matplotlib, seaborn
- Machine learning frameworks: PyTorch and libraries (Scikit-learn)
- Program/project management software: MS Project, Primavera

Strengths:

- Driven by challenge, ambitious, goal oriented, professional attitude, well-organized, self-motivated, and able to adapt rapidly to new environments.
- Experienced in leveraging scientific methods to inform qualitative and quantitative analyses
- Effective and comfortable in working across culturally diverse working environments.
- Expertise in building high achieving teams, project management, and solving complex problems through innovations and creativity.

Publications

Peer Reviewed Journals

- **Bulent Soykan**, Ghaith Rabadi: *A simulation-based optimization approach for multi-objective runway operations scheduling*, Simulation: Transactions of the Society for Modeling and Simulation International, June 2022. doi:10.1177/00375497221099544
- **Bulent Soykan**, Serpil Erol: *A Branch-and-Price Algorithm for the Robust Airline Crew Pairing Problem*, The Journal of Defense Sciences, Volume 13, Issue 1, 37-74, May 2014.
- **Bulent Soykan**: *C2 Agility? What? Why? How?*, The Transformers Spring/Summer 2015.

Peer Reviewed Book Chapters

- **Bulent Soykan**, Serpil Erol: *An Optimization-Based Decision Support Framework for Robust Airline Crew Pairing Process*. Using Decision Support Systems for Transportation Planning Efficiency, Edited by Ebru V. Ocalir-Akunal, 08/2015: chapter 1; IGI Global., ISBN: 9781466686489.
- **Bulent Soykan**, Ghaith Rabadi: *A Tabu Search Algorithm for the Multiple Runway Aircraft Scheduling Problem*. Heuristics, Metaheuristics and Approximate Methods in Planning and Scheduling, International Series in Operations Research & Management Science edited by Ghaith Rabadi, 01/2016: chapter 9; Springer International., ISBN: 978-3-319-26022-8.

Peer Reviewed Conference Papers

- Taner Altinok, Bekir Kazandir, **Bulent Soykan**, Filiz Ersoz*: Cost-Efficiency Analysis in Training Simulators: Example of Mortar Training Simulator and a Model Proposal*, METU, USMOS 2007: National Defense Applications in Modeling and Simulation Conference.
- **Bulent Soykan**, Ghaith Rabadi: *A Hybrid Metaheuristic Algorithm for Multi-Objective Runway Scheduling Problem in Simulation-based Optimization*. MODSIM World 2016, Virginia Beach/VA; 04/2016

- **Bulent Soykan**, Serpil Erol: *Airline Operational Scheduling Process*. The 3rd International Aviation Management Conference, Ankara, Turkey; 04/2016
- **Bulent Soykan**, Serpil Erol: *Simülasyon Deneyleri Tasarımı: Bir Simülasyon Modeli Üzerinde Karşılaştırmalı Çalışmalar / Designing Simulation Experiments: Comparative Studies on a Simulation Model*. USMOS 2015, ODTU, Ankara / Turkey; 11/2015
- **Bulent Soykan**, David S Alberts: *Moving C2 Agility from a theory to a NATO Practice*. 20th International Command and Control Research and Technology Symposium; 07/2015
- Mustafa Canan, **Bulent Soykan**: *Shared situational awareness/understanding: A key attribute of mission command concept for NATO*. 21st International Command and Control Research and Technology Symposium; 09/2016
- **Bulent Soykan**, Serpil Erol: *Analysis of Flight Delay Propagation with Bayesian Networks*, International Conference on Engineering Technologies ICENTE 2020; 11/2020
- **Bulent Soykan**, Fatih Kasimoglu: *Comparative studies on designing simulation experiments*, ICENTE 2020; 11/2020
- Fatih Kasimoglu, **Bulent Soykan**: *Worst case risk analysis of a transshipment network*, International Conference on Engineering Technologies ICENTE 2020; 11/2020
- **Bulent Soykan**, Serpil Erol: *Robust Scheduling and Schedule Optimization Approaches That Consider Operational Uncertainty*, YONAR2021, 04/2021
- **Bulent Soykan**, Serpil Erol: *Airline Scheduling Process Modeling Approaches and Solution Techniques*, YONAR2021, 04/2021

Technical Reports

- **Bulent Soykan**, Ghaith Rabadi: *Multi-Objective Simulation-based Optimization of Runway Operations Scheduling Using a Hybrid Metaheuristic Algorithm*. Technical Report for Airport Cooperative Research Program, 2016
- Micheline Belanger, Bjorn Johansson, David Alberts, **Bulent Soykan**, et al.: *C2 Agility Next Steps*, NATO Science and Technology Organization System Analysis and Studies Panel, 2018. ISBN 978-92-837-2171-0

Professional Certifications

- Project Management Professional (**PMP**) Certificate (2037599)
- **PRINCE2 Foundation** & **Practitioner** Certificate in Project Management (02317091-02-QFNY)
- **MSP Foundation** Certificate in Programme Management (GR636001987BS)
- **MSP Practitioner** Certificate in Programme Management (GR637001208BS)
- **ITIL v3 Foundation** Certificate in IT Service Management (02311837-01-WVF8)

Selected Projects

- **AI Solutions for Business Operations – 2020-2021**
  -Conceptualized and wrote code for optimization models to enhance business processes
  -Worked together with Software Engineers to make these models as robust as possible
  -Actively participated in business analytics and data science community
- **Terrorist Attack Pattern Modeling, Simulation & Analysis – 2016-2018**
  -Designed and implemented an automated algorithm that rank clusters of activity based on temporal and spatial metrics to examine the relationship between events in a series
  -Integrated geospatial and statistical analyses which enabled analysts to identify patterns as soon as they became apparent and improved the detection portion of the targeting cycle
- **Airline Crew Scheduling – 2011-2015**
  -Designed and implemented a discrete-event simulation model in C++ from scratch
  -Developed and implemented several meta-heuristic algorithms in Python to help automate scenario development and to be used as a research tool
- **Simulation Supported Wargaming for the Planning of Military Operations – 2009-2013**
  -Integrated agent-based simulation supported wargaming in the planning process which provide a convenient way to wargame and analyze several courses of action during analysis of plans and thereby increase the quality of plans and decrease the planning time
  -Designed and implemented a wargaming tool as a versatile simulation model capable of supporting an agent-based simulation of conflict at multiple levels of fidelity
  -Demonstrated the tool for the prospective users and evaluated/validated its effects on the planning process through experiments

- **Mission Command for Future Dismounted Soldier System – 2009-2013**
  -Developed an assessment framework to inform senior staff and established robust database systems to better facilitate the analysis of assessments
  -Utilized agent-based modeling to focus on C2 structures, processes, and mechanisms for adaptation instead of older systems analysis which were structured around units and platforms
  -Proactively worked with key stakeholders to gain domain expertise and used real-world data to ensure that the most salient aspects of the capability are reflected in the analysis

Teaching Experience

- **EM511 Optimization Methods,** Altinbas University, Graduate, 2020-2021
- **END321 System Simulation,** Altinbas University, Undergrad, 2020-2021
- **IES100 Introduction to Engineering Sciences,** Altinbas University, Undergrad, 2020-2021
- **END202 Statistics for Engineers,** Altinbas University, Undergrad, 2020-2021
- **END475 Engineering & Technology Management,** Altinbas University, Undergrad, 2020-2021
- **IES100 Introduction to Engineering Sciences,** Altinbas University, Undergrad, 2021-2022
- **IE491 Industrial Engineering Design Project I,** Altinbas University, Undergrad, 2021-2022
- **IE491 Industrial Engineering Design Project II,** Altinbas University, Undergrad, 2021-2022
- **END210 Systems Analysis & Design,** Istanbul Gedik University, Undergrad, 2021-2022
- **END364/END368 Simulation,** Istanbul Gedik University, Undergrad, 2021-2022

Selected Professional Trainings & Courses

- NATO Strategic Writing Course (NATO HQ SACT)
- NATO Staff Officer Orientation Course (NATO School Oberammergau)
- NATO Consultation, Command and Control (C3) Course (NATO School Oberammergau)
- NATO Academic Instructor Course (NATO School Oberammergau)
- Special Ops Component Command Staff Course (NATO Special Ops School Belgium)
- ACT Programme and Project Management Training (NATO HQ SACT)
- SOW Writing and COTR Training Program for NATO ACT (The George Washington University Law School, Washington, D.C.)
- Scheduling and Cost Control (The George Washington University School of Business)
- Program Management (The George Washington University School of Business)
- Project Leadership, Management and Communications Course (ESI Intl. Virginia)
- Managing Projects (ESI International Virginia)
- Business Process Analysis, Innovation and Design (ESI Intl. Virginia)
- Negotiation Skills for Project Managers (ESI Intl. Virginia)
- Requirements Management: A Key to Project Success (ESI Intl. Virginia)
- Project Management Professional (PMP) Training (Infotec Virginia)
- Deploying Operations Research/Systems Analysis (ORSA) Analyst (NATO HQ SACT)
- Machine Learning, Data Science and Deep Learning with Python (Udemy)
- Artificial Intelligence: Reinforcement Learning in Python (Udemy)

Awards & Honors

- **The second-place winner** in Airport Cooperative Research Program University Design Competition for Addressing Airport Needs in the category of Airport Management and Planning, funded by the FAA, USA, June 2016
- Received **2 NATO Service Medals** (SFOR and ISAF), numerous honors & letters of commendation, and decorations during his military career

References

- Ghaith Rabadi, Ph.D., Professor and Graduate Programs Director, School of Modeling, Simulation, and Training, University of Central Florida, Orlando / FL / USA, Ghaith.Rabadi@ucf.edu
- Serpil Erol, Ph.D., Professor, Gazi University, Department of Industrial Engineering, Ankara / Turkey, serpiler@gazi.edu.tr
- Mehmet Kinaci, Ph.D., Strategic Analyst, NATO HQ SACT, Norfolk / VA / USA, Mehmet.Kinaci@act.nato.int